

Green Fleet Optimization Report

PS-138 — Quantum-Inspired Fuel Consumption Prediction and Green Fleet Optimization

1. Executive Summary

A fleet of 14 vessels was evaluated against 3 route demands using the **QGA** optimiser. The recommended plan consumes **2,986 t** of fuel at a total cost of **\$3,789,751**, emitting **5,554 t CO2e** on a lifecycle basis, while meeting **100.0%** of cargo demand at **100.0%** schedule reliability. Constraint status: **feasible**.

2. Fleet Information

Vessel class	Count	Total DWT	Mean DWT
Capesize	3	379,200	126,400
Panamax	5	446,446	89,289
Supramax	4	200,499	50,125
Handysize	2	52,999	26,500

3. Input Scenario

Route	Distance (nm)	Cargo demand (t)	Deadline (h)	Weather
Port Hedland - Qingdao	3,600	240,000	340	MODERATE
Tubarao - Rotterdam	5,100	150,000	470	ROUGH
Paradip - Singapore	2,100	90,000	215	CALM

Carbon price \$0/t. Emission ceiling: none. Maximum tolerated delay 24 h.

4. Fuel Consumption Prediction

ML model **GradientBoosting** predicted **371.4 t** for the reference voyage (95% interval 198.3 to 544.5 t).

Model	MAE (t)	RMSE (t)	R2	MAPE (%)
RandomForest	67.46	118.98	0.9479	11.99
GradientBoosting	46.19	88.32	0.9713	7.79

Promoted model: **GradientBoosting**, selected by held-out RMSE on 1,200 unseen voyages.

5. Optimization Methodology

Discrete decisions (vessel deployment, route assignment, fuel selection) are solved with a Quantum-Inspired Genetic Algorithm in which each gene is a vector of probability amplitudes evolved by a quantum rotation gate and collapsed by observation. Continuous decisions (cruising speed per vessel) are solved with Quantum-Inspired Particle Swarm Optimization, which replaces velocity with a delta-potential-well position collapse. Both execute on classical hardware; no quantum computer is used at any point.

6. Selected Algorithm

QGA, runtime 0.41 s.

7. Optimization Results

Vessel	Class	Route	Cargo t	kn	Fuel	t	Cost \$	CO2e t	ETA h	Util %
V001	Capesize	Tubarao - Rotterda	100,000	11.3	MGO	504	640,081	1,932	493	100
V003	Panamax	Port Hedland - Qin	90,899	11.0	LNG	274	318,088	972	357	100
V004	Supramax	Paradip - Singapor	42,750	11.9	AMMONIA	251	353,790	55	200	100
V005	Supramax	Port Hedland - Qin	52,000	10.9	HFO	229	236,319	847	362	100
V007	Capesize	Port Hedland - Qin	97,101	10.9	LNG	294	390,841	1,045	360	77
V008	Panamax	Tubarao - Rotterda	50,000	11.9	AMMONIA	1,138	1,496,503	250	469	58
V011	Supramax	Paradip - Singapor	47,250	12.2	METHANOL	296	354,128	452	197	82

7 vessels deployed, 7 held idle.

8-10. Fuel, Cost and Emission Analysis

Metric	Baseline	Optimized	Change %	Direction
Fuel consumption (t)	2,582.8	2,985.9	+15.60%	worse
Operating cost (USD)	2,142,144.3	3,789,750.9	+76.91%	worse
Lifecycle CO2e (t)	9,577.2	5,554.2	-42.01%	improved
Cargo fulfilment (%)	100.0	100.0	+0.00%	worse
Schedule reliability (%)	100.0	100.0	+0.00%	worse

Percentages are computed from the two evaluations above; none are assumed.

11. Pareto Analysis

ID	Cost weight	Cost \$	Lifecycle CO2e t	Cargo %	Pareto-optimal
S00	0.00	3,548,358	7,047	100	no
S01	0.50	1,991,245	6,349	100	yes
S02	1.00	2,230,647	7,771	100	no

1 of 3 sampled configurations are non-dominated. Method: Weighted-sum scalarisation sweep with non-dominated filtering.

12. Benchmarking Results

Algorithm	Runs	Mean fitness	Std dev	Best	Worst	Iter. to converge	Runtime s
QGA	1	1.0265	0.0000	1.0265	1.0265	27.0	0.67
GA	1	1.0851	0.0000	1.0851	1.0851	28.0	0.09
QPSO	1	0.9159	0.0000	0.9159	0.9159	18.0	0.12
PSO	1	0.9156	0.0000	0.9156	0.9156	14.0	0.11
GREEDY	1	1.0386	0.0000	1.0386	1.0386	-	0.00

QGA vs GA: better mean fitness achieved by **QGA** (gap -5.40%).

QPSO vs PSO: better mean fitness achieved by **PSO** (gap +0.04%).

Independent runs with distinct random seeds. Mean and standard deviation are reported because metaheuristic outcomes vary by run. No result is asserted in advance of the experiment.

13. Alternative Fuel Comparison

Fuel	\$/t	Annual t	Annual cost \$	Lifecycle CO ₂ e t	vs base %	Payback y
HFO	520	7,478	3,888,632	27,729	+0.0	n/a
MGO	720	7,037	5,066,588	26,994	-2.6	n/a
LNG	640	6,267	4,010,675	22,247	-19.8	n/a
METHANOL	980	15,106	14,803,723	23,036	-16.9	n/a
AMMONIA	1,150	16,160	18,584,296	3,555	-87.2	n/a
HYDROGEN	4,200	2,505	10,521,741	1,127	-95.9	n/a

Transport work is held constant across fuels. Capex and prices are configurable demo parameters, not quotations.

14. Compliance Analysis

Target	Required %	Achieved %	Current g/t-nm	Target g/t-nm	Status
IMO_2030	40	69.7	2.363	4.680	Compliant
IMO_2040	70	69.7	2.363	2.340	Attention Required
IMO_2050	100	69.7	2.363	0.000	Above Target

EU ETS estimate for 2026: 2,148 t of allowances at a 100% phase-in and 50% voyage coverage, costing approximately \$0.

Model estimate from configurable parameters. Not a regulatory calculation, certification or legal advice. Actual compliance depends on audited operational data and the rules in force for the relevant compliance year.

15. Recommendations

1. The optimised plan consumes 15.6% MORE fuel than the baseline. Re-check the objective weights before acting on it.
2. Emission reduction of 42.0% is achievable without reducing cargo fulfilment below 100%.
3. The selected plan satisfies every declared constraint and is operationally feasible.
4. On this scenario over 1 independent runs, QGA achieved the better mean fitness in QGA vs GA (gap 5.40%).

- 5. On this scenario over 1 independent runs, PSO achieved the better mean fitness in QPSO vs PSO (gap 0.04%).
- 6. No alternative fuel pays back within the modelled horizon at current prices and carbon cost. Revisit if the carbon price rises materially.
- 7. IMO 2040 - indicative checkpoint: currently 69.7% reduced against a required 70%. Closing the gap needs a 1% intensity improvement.

16. Methodology Notes

Fuel consumption follows a cubic speed-power relationship with a load-dependent specific fuel oil consumption curve minimised near 75% MCR, corrected for cargo displacement, hull fouling with age, and added resistance from wind, waves and current. Emissions are split well-to-tank and tank-to-wake; lifecycle CO2e is their sum. The optimiser fitness is a weighted sum of normalised fuel, cost, emission and reliability terms plus a static penalty on normalised constraint violation.

17. Dataset Information

Demo dataset generated for simulation and algorithm validation. 6,000 synthetic voyages produced from the documented physical model with injected flow-meter, per-hull and weather noise. This is not real shipping-company data and must not be presented as such.

18. Report Metadata

Generated	2026-09-29 23:33:00
Platform	GreenFleet Quantum
Problem statement	PS-138
Scenario	Case Study 1 - Conventional Fleet
Algorithm	QGA
Compute	Classical hardware; quantum-inspired algorithms only